阿思丹 | ASDAN China

MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes
考试时长：75 分钟
Language: English & Chinese
试卷语言：英文和中文

This is a 30-question multiple-choice competition.
考试包含 30 道单项选择题。
Problems 1 to 10 are worth 3 points each.
1-10 题每题 3 分
Problems 11 to 20 are worth 4 points each.
11-20 题每题 4 分
Problems 21 to 30 are worth 5 points each.
21-30 题每题 5 分
1 point will be deducted for an incorrect answer and no point for a blank answer.
答错扣 1 分，不答不扣分。
The full score is 150 (Starting point is 30)
总分为 150 分（起始分数为 30 分）

Do not open this booklet until you are told to do so.
收到监考老师指令后，方可打开试卷册。
Mark your answers clearly on the answer sheet using a **2B pencil**.
务必使用 **2B 铅笔**在答题卡上正确填涂自己的答案。
Calculator cannot be used during the competition.
考试期间禁止使用计算器。

Good Luck!
预祝考试顺利

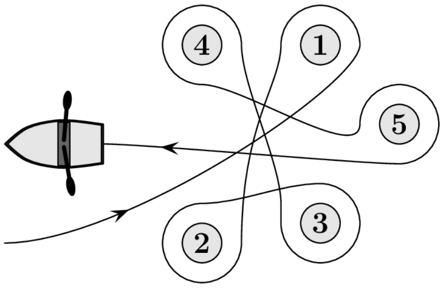
Level 4

Year 7 & Year 8

3 points

1. Meike paddled around five buoys, as shown. Around which of the buoys did Meike paddle in a clockwise direction?

如图所示，Meike 绕着五个浮标划船。请问他绕着哪几个浮标划船时是顺时针方向?



(A) 2, 3 and 4　　2, 3 和 4

(B) 1, 2 and 3　　1, 2 和 3

(C) 1, 3 and 5　　1, 3 和 5

(D) 2, 4 and 5　　2, 4 和 5

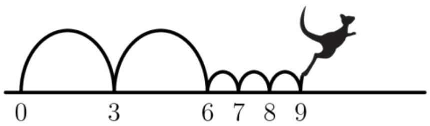
(E) 2, 3 and 5　　2, 3 和 5

2. Beate rearranges the five numbered pieces shown to display the smallest possible nine-digit number. Which piece does she place at the right-hand end?

Beate 打算重新排列以下五个数字块，使得到的九位数尽可能小。请问她应该把哪一块放在最右边?

(A) $\boxed{4}$　　(B) $\boxed{8}$　　(C) $\boxed{31}$　　(D) $\boxed{59}$　　(E) $\boxed{107}$

3. Kengu enjoys jumping on the number line. He always makes two large jumps followed by three small jumps, as shown, and then repeats this process over and over again. Kengu starts his jumping routine on 0. On which of these numbers will Kengu land during his routine?



Kengu 喜欢在数轴上跳跃。如图所示，他总是先两大跳，再三小跳，然后不断重复这个过程。现在 Kengu 从 0 开始跳，请问整个跳跃过程中它会跳到以下哪个数字上?

(A) 82　　(B) 83　　(C) 84　　(D) 85　　(E) 86

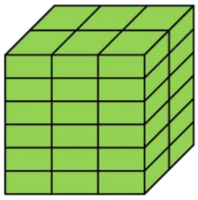
4. The number plate of Kangy’s car fell off. He put it back upside down but luckily this didn’t make any difference. Which one of the following could be Kangy’s number plate?

Kangy 的车牌掉了，但他重装的时候上下颠倒了。所幸的是，与原先并无差别。以下哪一个可能是 Kangy 的车牌?

(A) **04 NSN 40**　　(B) **60 HOH 09**　　(C) **80 BNB 08**

(D) **03 HNH 30**　　(E) **08 XBX 80**

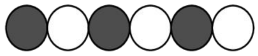
5. Rob the Builder has a brick whose shortest side is 4 cm. He uses several such bricks to build the cube shown. What are the dimensions, in cm, of his brick?



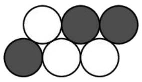
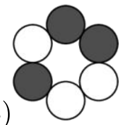
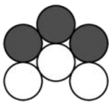
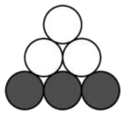
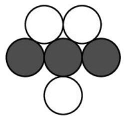
建筑工人 Rob 有一个短边长为 4 厘米的小砖块。于是他又找了一些这样的小砖块建造成如右图所示的正方体。请问这个小砖块的尺寸是多少（单位：厘米）？

(A) 4×6×12　(B) 4×6×16　(C) 4×8×12　(D) 4×8×16　(E) 4×12×16

6. The black and white caterpillar shown in the picture curls up to sleep. Which of the following could be seen?



如图是一只黑白相间的毛毛虫，它正蜷缩着睡觉。请问可能是以下哪个形状?

(A)  (B)  (C)  (D)  (E) 

7. In the statement below there are five empty spaces. Sanja wants to fill four of them with plus signs and one with a minus sign so that the statement is correct. Where should she place the minus sign?

$$6\square 9\square 12\square 15\square 18\square 21=45$$

图中的等式有 5 个空格。Sanja 想用 4 个加号和 1 个减号使等式成立，请问她应该把减号填在哪里?

(A) Between 6 and 9　6 和 9 之间

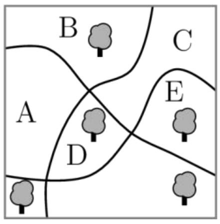
(B) Between 9 and 12　9 和 12 之间

(C) Between 12 and 15　12 和 15 之间

(D) Between 15 and 18　15 和 18 之间

(E) Between 18 and 21　18 和 21 之间

8. There are five big trees and three paths in a park. In which region of the park should a new tree be planted so that for each path, there are the same number of trees on both sides?



公园里有五棵大树和三条小路。请问在公园的哪个区域再种一棵树，可以使得每条小路两侧的树木总数相同?

(A) A　(B) B　(C) C　(D) D　(E) E

9. How many positive integers between 100 and 300 have only odd digits?
请问 100 到 300 之间有多少个正整数的数位上只有奇数?

(A) 25　(B) 50　(C) 75　(D) 100　(E) 150

10. Gerard wrote down the sum of squares of two numbers, as shown. Unfortunately some of the digits cannot be seen because they are covered in ink. What is the last digit of the first number?

如图所示，Gerard 写下了两个数的平方和。可惜的是，个别数字被墨水盖住了，无法辨认出来。请问第一个数的最后一位数是几？

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

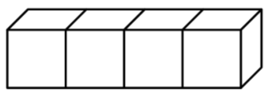
4 points

11. The distance between two shelves in the cupboard in Monica's kitchen is 36 cm. She knows that a stack of 8 of her favourite glasses is 42 cm tall and that a stack of 2 glasses is 18 cm tall. What is the largest number of glasses that can be stacked and still fit onto a shelf?

Monica 的厨房橱柜中两个搁板之间的距离是 36 厘米。她最喜欢的杯子如果 8 个叠成一摞高度为 42 厘米，如果 2 个叠成一摞高度为 18 厘米。请问搁板上最多可以叠放多少个玻璃杯？

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

12. On a standard dice, the sum of the numbers of dots on opposite faces is always 7. Four standard dice are glued together, as shown. What is the minimum number of dots that could lie on the whole surface?



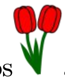
一枚标准骰子相对面上的点数之和总是 7。如图所示，将四枚标准骰子粘在一起，请问整个表面上的最小点数之和是多少？

(A) 52 (B) 54 (C) 56 (D) 58 (E) 60

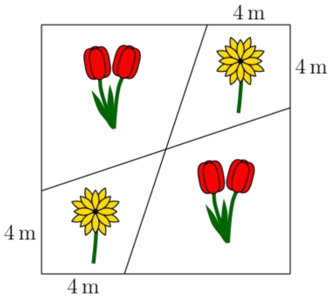
13. Three sisters, whose average age is 10, each have different ages. When they get together in pairs, the average ages of two such pairs are 11 and 12. What is the age of the eldest sister?

三姐妹年龄各不相同，她们的平均年龄为 10 岁。当她们两两为一组时，有两组的平均年龄是 11 岁和 12 岁。请问最大的姐姐是多少岁？

(A) 10 (B) 11 (C) 12 (D) 14 (E) 16

14. Tony the Gardener planted tulips  and daisies  in a square flowerbed with side-length 12m, arranged as shown. What is the total area of the regions in which he planted daisies?

园丁Tony 在边长为 12 米的方形花坛上种植了郁金香  和雏菊  ，位置如图所示。请问他种植雏菊的区域总面积是多少?



(A) $48m^2$ 48 平方米 (B) $46m^2$ 46 平方米 (C) $44m^2$ 44 平方米

(D) $40m^2$ 40 平方米 (E) $36m^2$ 36 平方米

15. In my office, there are two clocks. One clock gains one minute every hour and the other loses two minutes every hour. Yesterday I set them both to the correct time but when I looked at them today, I saw that the time shown on one was 11:00 and shown on the other was 12:00. What time was it when I set the two clocks?

我的办公室有两只时钟。一只钟每小时会快一分钟，另一只钟每小时会慢两分钟。昨天我把它们都调到了正确时间。我今天看时间时，一只显示 11:00，另一只显示 12:00。请问我昨天调它们的时候是几点?

(A) 23:00 (B) 19:40 (C) 15:40 (D) 14:00 (E) 11:20

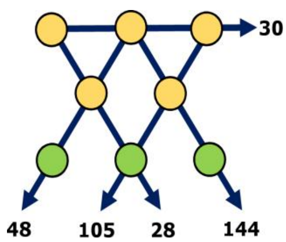
16. Werner wrote several positive numbers smaller than 7 on a piece of paper. Ria then crossed out all Werner's numbers and replaced each of them with their difference from 7. The sum of Werner's numbers was 22. The sum of Ria's numbers is 34. How many numbers did Werner write down?

Werner 在纸上写了几个小于 7 的正数。然后 Ria 划掉了 Werner 写的所有数字，并用这些数字与 7 的差值将其替换。Werner 写的数字之和是 22，Ria 写的数字之和是 34，请问 Werner 写了多少个数字?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

17. The numbers 1 to 8 are placed, once each, in the circles shown. The numbers by the arrows show the products of the three numbers in the circles on that straight line. What is the sum of the numbers in the three circles at the bottom of the figure?

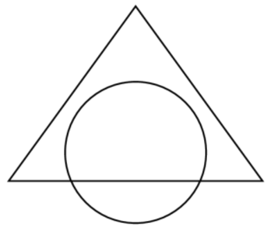
将数字 1 到 8 放置在如图所示的圆圈中，每个数字只能使用一次。箭头指向的数字表示箭头所在直线上的三个数字之积。请问图中底部三个圆圈中的数字之和是多少?



(A) 11 (B) 12 (C) 15 (D) 17 (E) 19

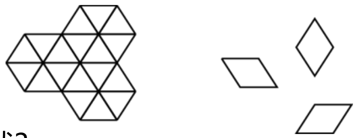
18. The area of the intersection of a circle and a triangle is 45% of the area of their union. The area of the triangle outside the circle is 40% of the area of their union. What percentage of the circle lies outside the triangle?

如图所示，圆和三角形重叠部分的面积占整个图形面积的 45%。位于圆外的三角形面积占整个图形面积的 40%。请问圆在三角形外的面积占圆面积的百分比是多少？



(A) 20%　(B) 25%　(C) 30%　(D) 35%　(E) 50%

19. In how many ways can the shape on the left be completely covered using nine tiles like the ones on the right?



一共有多少种方式可以用九块右边的小瓷砖恰好完全覆盖住左边的形状？

(A) 1　(B) 6　(C) 8　(D) 9　(E) 12

20. Marc always cycles at the same speed and he always walks at the same speed. He can cover the round trip from his home to school and back again in 20 minutes when he cycles and in 60 minutes when he walks. Yesterday Marc started cycling to school but stopped and left his bike at Eva’s house on the way before finishing his journey on foot. On the way back, he walked to Eva’s house, collected his bike and then cycled the rest of the way home. His total travel time was 52 minutes. What fraction of his journey did Marc make by bike?

Marc 总是以相同的速度骑自行车，以相同的速度走路。他可以在 20 分钟内骑车完成从家到学校的往返路程；也可以在 60 分钟内步行完成往返。昨天，Marc 开始骑自行车上学，但他中途停下，把自行车留在了Eva 家，然后步行到学校。放学回家时先步行到 Eva 家取回自行车，然后骑车回家。他全程共花费了 52 分钟。请问 Marc 骑自行车所走的路程占总路程的比例是多少？

(A) $\frac{1}{6}$　(B) $\frac{1}{5}$　(C) $\frac{1}{4}$　(D) $\frac{1}{3}$　(E) $\frac{1}{2}$

5 points

21. Jenny decided to enter numbers into the cells of a 3 × 3 table so that the sum of the numbers in all four possible 2 × 2 squares will be the same. The numbers in three of the corner cells have already been written, as shown. Which number should she write in the fourth corner cell?

2		4
?		3

Jenny 打算在 3 × 3 的方格中填入数字，使所有可能的 2 × 2 小方格中的四个数字之和都相同。如图所示，三个拐角单元格中已填好数字，请问第四个拐角单元格中的数字应该填几？

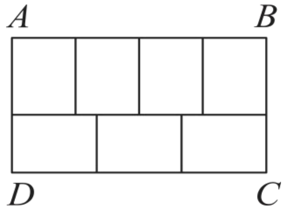
(A) 0　(B) 1　(C) 4　(D) 5　(E) 6

22. The villages A, B, C and D are situated, not necessarily in that order, on a long straight road. The distance from A to C is 75 km, the distance from B to D is 45 km and the distance from B to C is 20 km. Which of the following could not be the distance from A to D?

村庄 A、B、C 和 D 坐落在一条又长又直的道路上，但不一定按照这个顺序排列。从 A 到 C 的距离是 75 公里，从 B 到 D 的距离是 45 公里，从 B 到 C 的距离是 20 公里。下列哪项不可能是从 A 到 D 的距离？

(A) 10km 10公里 (B) 50km 50公里 (C) 80km 80公里
(D) 100km 100公里 (E) 140km 140公里

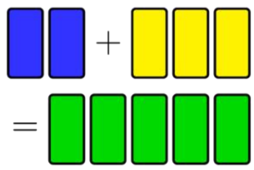
23. The large rectangle $ABCD$ is divided into seven identical rectangles. What is the ratio $\frac{AB}{BC}$?



大矩形 $ABCD$ 被分成七个相同的小矩形。请问 $\frac{AB}{BC}$ 等于多少？

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$ (C) $\frac{8}{5}$ (D) $\frac{12}{7}$ (E) $\frac{7}{3}$

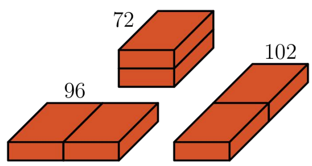
24. A painter wanted to mix 2 litres of blue paint with 3 litres of yellow paint to make 5 litres of green paint. However, by mistake he used 3 litres of blue and 2 litres of yellow so that he made the wrong shade of green. What is the smallest amount of this green paint that he must throw away so that, using the rest of his green paint and some extra blue and/or yellow paint, he could make 5 litres of paint of the correct shade of green?



一位画家想用 2 升蓝色颜料与 3 升黄色颜料混合制成 5 升绿色颜料。但他不小心用成了 3 升蓝色颜料和 2 升黄色颜料，导致最后制成的绿色颜料色调不对。请问他必须最少去掉多少绿色颜料，才能用剩下的绿色颜料和另外一些蓝色和/或黄色颜料来制出 5 升正确绿色色调的颜料？

(A) $\frac{5}{3}$ litres $\frac{5}{3}$ 升 (B) $\frac{3}{2}$ litres $\frac{3}{2}$ 升 (C) $\frac{2}{3}$ litres $\frac{2}{3}$ 升
(D) $\frac{3}{5}$ litres $\frac{3}{5}$ 升 (E) $\frac{5}{9}$ litres $\frac{5}{9}$ 升

25. A builder has two identical bricks. She places them side by side in three different ways, as shown. The surface areas of the three shapes obtained are 72, 96 and 102. What is the surface area of the original brick?

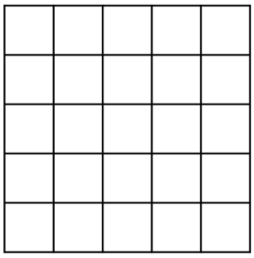


一名建筑工人有两块完全相同的砖。如图所示，她用三种不同的方式将它们并排放置，这三种形状砖块的表面积分别为 72、96 和 102。请问一块砖本身的表面积是多少？

(A) 36 (B) 48 (C) 52 (D) 54 (E) 60

26. What is the smallest number of cells that need to be coloured in a 5×5 square so that any 1×4 or 4×1 rectangle lying inside the square has at least one cell coloured?

在一个 5×5 的正方形中，要使正方形内任意一个 1×4 或 4×1 的小矩形中都至少有一个涂色单元格，请问最少需要将多少个单元格涂色?



(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

27. Mowgli asks a zebra and a panther what day it is. The zebra always lies on Monday, Tuesday and Wednesday. The panther always lies on Thursday, Friday and Saturday. The zebra says, "Yesterday was one of my lying days." The panther says "Yesterday was also one of my lying days." What day is it?

Mowgli 问斑马和黑豹今天是星期几。斑马总是在星期一、星期二和星期三说谎，黑豹总是在星期四、星期五和星期六说谎。斑马回答说："昨天我说谎了。" 黑豹说："昨天我也说谎了。"

请问今天是星期几?

(A) Thursday 星期四 (B) Friday 星期五 (C) Saturday 星期六

(D) Sunday 星期日 (E) Monday 星期一

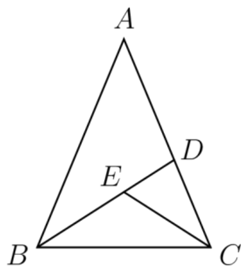
28. Several points are marked on a line. Renard then marked another point between each two adjacent points on the line. He repeated this process a further three times. There are now 225 points marked on the line. How many points were marked on the line initially?

一条线段上标有几个点，然后 Renard 在每对相邻点之间标记了另一个点。他又重复了三次这个过程。现在这条线段上有 225 个点。请问他最初在这条线段上标记了多少个点?

(A) 10 (B) 12 (C) 15 (D) 16 (E) 25

29. An isosceles triangle ABC, with $AB = AC$, is split into three smaller isosceles triangles, as shown, so that $AD = DB$, $CE = CD$, and $BE = EC$. Note that the diagram is not drawn to scale. What is the size, in degrees, of angle BAC?

等腰三角形 ABC 中，$AB = AC$，将它划分成三个较小的等腰三角形，如图所示，$AD = DB$，$CE = CD$，$BE = EC$。注意，该图未按比例绘制。请问 $\angle BAC$ 等于多少度?



(A) 24 (B) 28 (C) 30 (D) 35 (E) 36

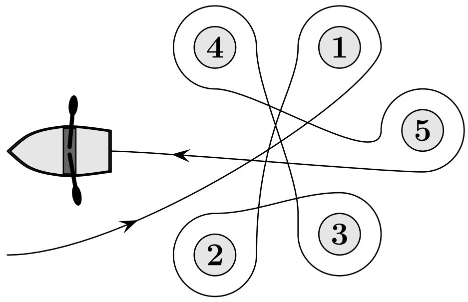
30. There are 2022 kangaroos and some koalas living across seven parks. In each park the number of kangaroos is equal to the total number of koalas in all the other parks. How many koalas live in the seven parks in total?

七个公园里共生活着 2022 只袋鼠和一些考拉。每个公园里的袋鼠数量等于其他所有公园的考拉数量之和。请问七个公园里一共有多少只考拉?

(A) 288 (B) 337 (C) 576 (D) 674 (E) 2022

3 points

1 (DE). Meike paddled around five buoys, as shown. Around which of the buoys did Meike paddle in a clockwise direction?



(**A**) 2, 3 and 4
(**B**) 1, 2 and 3
(**C**) 1, 3 and 5
(**D**) 2, 4 and 5
(**E**) 2, 3 and 5

SOLUTION: Follow the lines and add more arrows. Around buoys 2, 4 and 5 Meike paddes in a clockwise direction.

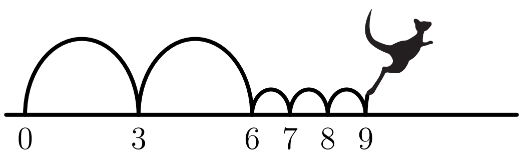
2 (DK). Beate rearranges the five numbered pieces shown to display the smallest possible nine-digit number.

Which piece does she place at the right-hand end?

(**A**) $\boxed{4}$ (**B**) $\boxed{8}$ (**C**) $\boxed{31}$ (**D**) $\boxed{59}$ (**E**) $\boxed{107}$

SOLUTION: The smallest possible number that can be made with the five pieces is: $\boxed{107}\,\boxed{31}\,\boxed{4}\,\boxed{59}\,\boxed{8}$

3 (NO). Kengu enjoys jumping on the number line. He always makes two large jumps followed by three small jumps, as shown, and then repeats this process over and over again. Kengu starts his jumping routine on 0. On which of these numbers will Kengu land during his routine?



(**A**) 82 (**B**) 83 (**C**) 84 (**D**) 85 (**E**) 86

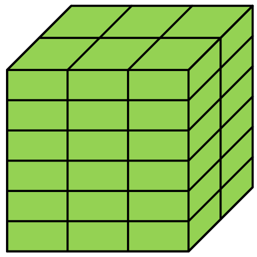
SOLUTION: During every routine Kengu advances 9 units. So after 9 routines he ends up at 81. After that he will land at 81+3 = 84. The next jump will bring him to 84+6 = 90, which is already to big.

4 (FR). The number plate of Kangy's car fell off. He put it back upside down but luckily this didn't make any difference. Which one of the following could be Kangy's number plate?

(**A**) **04 NSN 40** (**B**) **60 HOH 09** (**C**) **80 BNB 08**
(**D**) **03 HNH 30** (**E**) **08 XBX 80**

SOLUTION: Only B shows upside down the same number plate: **60 HOH 09**.

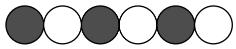
5 (GR). Rob the Builder has a brick whose shortest side is 4 cm. He uses several such bricks to build the cube shown. What are the dimensions, in cm, of his brick?

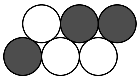
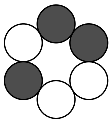
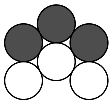
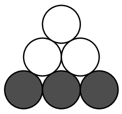
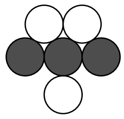


(**A**) $4 \times 6 \times 12$ (**B**) $4 \times 6 \times 16$ (**C**) $4 \times 8 \times 12$
(**D**) $4 \times 8 \times 16$ (**E**) $4 \times 12 \times 16$

Solution: If the dimensions of the brick are H, W and L (here H=4 cm), from the figure we conclude that the side of the cube is 6×4=24 cm. Since the sides of the cube are equal we conclude from the figure that 3L=24 cm and 2W=24 cm. So L=8 cm and W= 12 cm.

6 (DE). The black and white caterpillar shown in the picture curls up to sleep. Which of the following could be seen?



(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

Solution: Follow the form of the caterpillar by drawing a line. Only at (A) the line can be completed.

7 (IR). In the statement below there are five empty spaces. Sanja wants to fill four of them with plus signs and one with a minus sign so that the statement is correct.

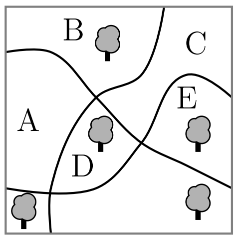
$$6\square 9\square 12\square 15\square 18\square 21=45$$

Where should she place the minus sign?

(**A**) Between 6 and 9 (**B**) Between 9 and 12 (**C**) Between 12 and 15
(**D**) Between 15 and 18 (**E**) Between 18 and 21

Solution: The total sum 6+9+12+15+18+21=81. This means that we counted 81-45=36 too much. So the minus sign should be at $36/2 = 18$. So 6+9+12+15-18+21=45.

8 (DE). There are five big trees and three paths in a park. In which region of the park should a new tree be planted so that for each path, there are the same number of trees on both sides?



(**A**) A (**B**) B
(**C**) C (**D**) D
(**E**) E

Solution: Look at the path from the bottom of the picture to the top: at the left hand side there are 2 trees and at the right hand side there are 3 trees. So, the new tree must be to the left of that path. Look at the path that starts high on the left hand side of the picture and ends low at the right hand side: above this path there are 2 trees and below this path there are 3 trees. So, the new tree must be above this path. So together with the previous observation, the new tree must be planted in region B.

9 (FR). How many positive integers between 100 and 300 have only odd digits?

(**A**) 25 (**B**) 50 (**C**) 75 (**D**) 100 (**E**) 150

Solution: The odd integers are 1,3,5,7, and 9. As a first digit we can only use '1'; as a second digit and as a third digit we can use all 5 possible odd integers. Hence the total number of integers between 100 and 300 with only odd digits is $1 \times 5 \times 5 = 25$.

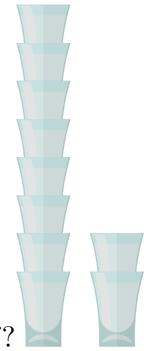
10 (GR). Gerard wrote down the sum of squares of two numbers, as shown. Unfortunately some of the digits cannot be seen because they are covered in ink. What is the last digit of the first number?

(**A**) 3 (**B**) 4 (**C**) 5 (**D**) 6 (**E**) 7

Solution: Consider the last digit only. As $(XXX2)^2$ and 7133029 end in 4 and 9 respectively, the last digit of the left most number is 5. To show that the situation is possible, consider $2385^2 + 1202^2 = 7133029$.

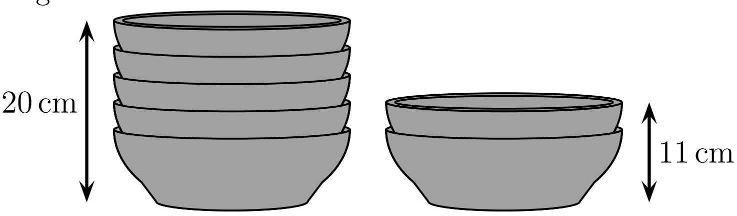
4 points

11 (NO). The distance between two shelves in the cupboard in Monica's kitchen is 36 cm. She knows that a stack of 8 of her favourite glasses is 42 cm tall and that a stack of 2 glasses is 18 cm tall. What is the largest number of glasses that can be stacked and still fit onto a shelf?



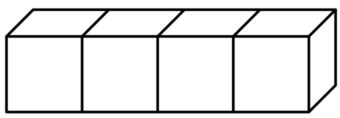
(**A**) 3 (**B**) 4 (**C**) 5 (**D**) 6 (**E**) 7

Solution: A stack of 8 glasses is 42 cm tall and a stack of 2 glasses is 18 cm tall, so for the extra 6 glasses measure 24 cm. This means that for every extra glass added to the stack, increases its height by $\frac{24}{6} = 4$ cm. If 4 glasses are added to the stack of 2 glasses it would create a stack of height 18 cm + 4 × 4 cm = 34 cm. This means that a stack of 6 glasses is the largest that can fit onto the shelf.
Alternative:
The distance between two shelves in the cupboard in Monika's kitchen is 30 cm. She knows that a stack of 5 of her favorite bowls is 20 cm tall and that a stack of 2 bowls is 11 cm tall. What is the largest number of bowls that can be stacked and still fit onto a shelf?



(A) 6/ (B) 7/ (C) 8/ (D) 9/ (E) 10 correct answer: C

12 (SE). On a standard dice, the sum of the numbers of dots on opposite faces is always 7. Four standard dice are glued together, as shown. What is the minimum number of dots that could lie on the whole surface?



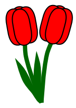
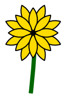
(**A**) 52 (**B**) 54 (**C**) 56 (**D**) 58 (**E**) 60

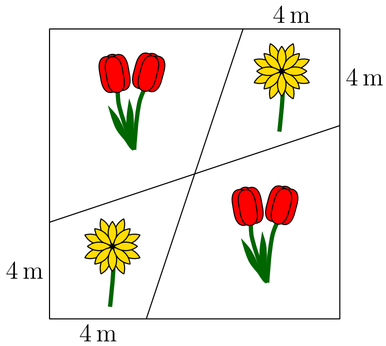
Solution: To have the minimum number of dots on the surface we want to glue together the faces with 6 dots of the two dice on the left and of the two dice on the right. This means that on both the left-hand side and on the right-hand side of the shape there is a 1 showing. The rest of the dots (2, 3, 4, and 5) are visible on the faces of each of the four dices. So the minimum numbers of dots on the surface is 4(2+3+4+5) + 2=58

13 (CL). Three sisters, whose average age is 10, each have different ages. When they get together in pairs, the average ages of two such pairs are 11 and 12. What is the age of the eldest sister?

(**A**) 10 (**B**) 11 (**C**) 12 (**D**) 14 (**E**) 16

Solution: Let a, b and c be the ages of the three sisters. Therefore, $\frac{a+b+c}{3} = 10$, $\frac{a+b}{2} = 11$ and $\frac{a+c}{2} = 12$. Hence $a+b+c = 30$, $a+b = 22$ and $a+c = 24$, and so $a = 16, b = 6, c = 8$. Therefore the eldest is 16.

14 (SK). Tony the Gardener planted tulips  and daisies  in a square flowerbed with side-length 12 m, arranged as shown. What is the total area of the regions in which he planted daisies?



(**A**) 48 m^2
(**B**) 46 m^2
(**C**) 44 m^2
(**D**) 40 m^2
(**E**) 36 m^2

Solution: It is obvious that the two quadrilaterals with daisies are of equal size. Divide each of the quadrilaterals into 2 triangles by connecting the vertex of the square with the intersection of the 2 lines.We obtain 4 equal triangles with area $\frac{1}{2} \cdot 4 \cdot 12m^2$. The total area of the regions with daisies therefore is $48m^2$.

15 (CT). In my office, there are two clocks. One clock gains one minute every hour and the other loses two minutes every hour. Yesterday I set them both to the correct time but when I looked at them today, I saw that the time shown on one was 11:00 and shown on the other was 12:00. What time was it when I set the two clocks?

(**A**) 23:00 (**B**) 19:40 (**C**) 15:40 (**D**) 14:00 (**E**) 11:20

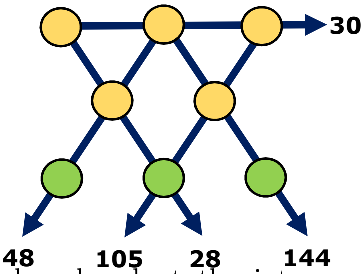
Solution: If one clock gains one minute every hour and the other loses two minutes every hour then the time difference increases by 3 minutes per hour. So if the time difference is 60 minutes, the last time that the clocks were set to the correct time is 20 hours ago. During that period, the fastest clock advanced an additional 20 minutes. So the clocks were set at 12:00 − 20:00 − 0:20 = 15:40.

16 (GR). Werner wrote several positive numbers numbers smaller than 7 on a piece of paper. Ria then crossed out all Werner's numbers and replaced each of them with their difference from 7. The sum of Werner's numbers was 22. The sum of Ria's numbers is 34. How many numbers did Werner write down?

(**A**) 7 (**B**) 8 (**C**) 9 (**D**) 10 (**E**) 11

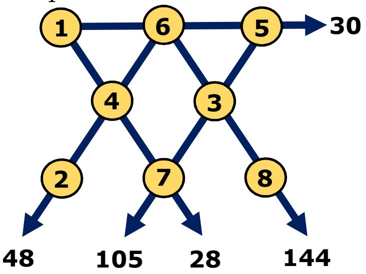
Solution: The sum of all Werner's numbers and all Ria's numbers together is 22+34=56. But when taken in pairs, Werner's number plus Ria's corresponding number, the sum of each pair is N+(7-N)=7. So there are 56:7=8 pairs. So originally 8 numbers were written on the paper

17 (GR). The numbers 1 to 8 are placed, once each, in the circles shown. The numbers by the arrows show the products of the three numbers in the circles on that straight line. What is the sum of the numbers in the three circles at the bottom of the figure?

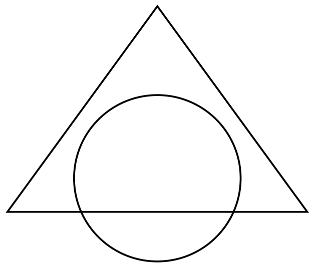


(**A**) 11 (**B**) 12 (**C**) 15 (**D**) 17 (**E**) 19

SOLUTION: The numbers 30 and 105 are multiples of 5, so 5 must be placed at the intersection of the corresponding lines. Similarly, the numbers 105 and 28 are multiples of 7, so 7 must be placed at the intersection of the corresponding lines. It is now easy to complete the figure.



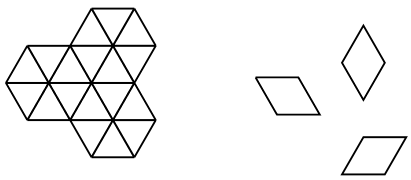
18 (UA). The area of the intersection of a circle and a triangle is 45% of the area of their union. The area of the triangle outside the circle is 40% of the area of their union. What percentage of the circle lies outside the triangle?



(**A**) 20% (**B**) 25%
(**C**) 30% (**D**) 35%
(**E**) 50%

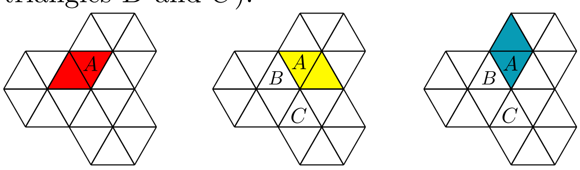
SOLUTION: Suppose the area of the union is 100. Then the area of the intersection is 45, and the area of the triangle outside the circle is 40. Hence, the area of the circle outside the triangle is $100 - 45 = 15$. Consequently, the percentage of the circle that lies outside the triangle is $\frac{15}{15+45} = 25\%$
Note: If the words intersection and union are not known then you could reword the question replacing intersection with "the overlapping region in both shapes" and union with "the whole figure".

19 (MX). In how many ways can the shape on the left be completely covered using nine tiles like the ones on the right?



(**A**) 1 (**B**) 6 (**C**) 8
(**D**) 9 (**E**) 12

SOLUTION: There are 3 ways to cover the triangle marked with A. The first one allows only 1 possibility; both the second and the third have 4 possibilities each (depending of the ways to cover triangles B and C).



20 (IT). Marc always cycles at the same speed and he always walks at the same speed. He can cover the round trip from his home to school and back again in 20 minutes when he cycles and in 60 minutes when he walks. Yesterday Marc started cycling to school but stopped and left his bike at Eva's house on the way before finishing his journey on foot. On the way back, he walked to Eva's house, collected his bike and then cycled the rest of the way home. His total travel time was 52 minutes. What fraction of his journey did Marc make by bike?

(**A**) $\frac{1}{6}$ (**B**) $\frac{1}{5}$ (**C**) $\frac{1}{4}$ (**D**) $\frac{1}{3}$ (**E**) $\frac{1}{2}$

SOLUTION: Let the fraction of the route that Marc did by bike be b. Therefore Marc rides his bike for $20b$ minutes and he walks for $60(1-b)$ minutes. Hence, $52 = 20b + 60(1-b)$, which gives $b = \frac{1}{5}$.

5 points

21 (BY). Jenny decided to enter numbers into the cells of a 3×3 table so that the sum of the numbers in all four possible 2×2 squares will be the same. The numbers in three of the corner cells have already been written, as shown. Which number should she write in the fourth corner cell?

2		4
?		3

(**A**) 0 (**B**) 1 (**C**) 4 (**D**) 5 (**E**) 6

SOLUTION:

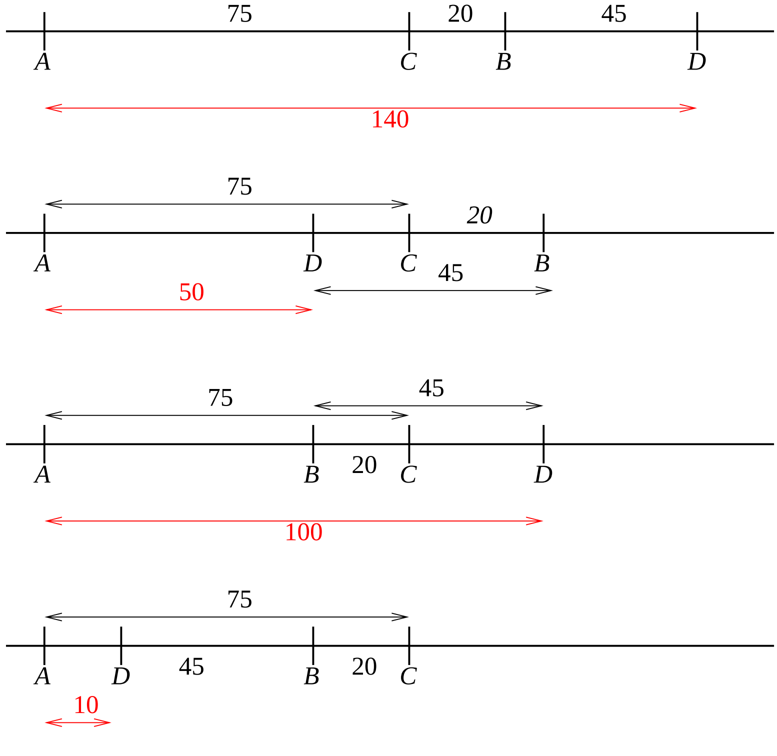
2		4
a		b
?		3

Note that $2 + a = 4 + b$, and that $? + a = 3 + b$. So, $? = 1$.

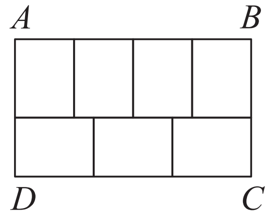
22 (PL). The villages A, B, C and D are situated, not necessarily in that order, on a long straight road. The distance from A to C is 75 km, the distance from B to D is 45 km and the distance from B to C is 20 km. Which of the following could not be the distance from A to D?

(**A**) 10 km (**B**) 50 km (**C**) 80 km (**D**) 100 km (**E**) 140 km

SOLUTION: All possible villages layouts are shown in the figure below.



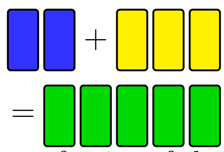
23 (RU). The large rectangle $ABCD$ is divided into seven identical rectangles. What is the ratio $\dfrac{AB}{BC}$?



(**A**) $\frac{1}{2}$ (**B**) $\frac{4}{3}$ (**C**) $\frac{8}{5}$ (**D**) $\frac{12}{7}$ (**E**) $\frac{7}{3}$

SOLUTION: Let the width of a small rectangle be w and the length of a small rectangle be l. Then, $4w = 3l$ or $l = \frac{4}{3}w$. Therefore, the ratio $\frac{AB}{BC} = \frac{4w}{\frac{4}{3}w+w} = \frac{4}{\frac{7}{3}} = \frac{12}{7}$.

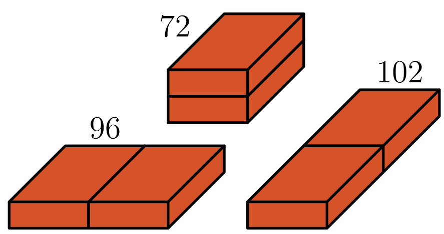
24 (GR). A painter wanted to mix 2 litres of blue paint with 3 litres of yellow paint to make 5 litres of green paint. However, by mistake he used 3 litres of blue and 2 litres of yellow so that he made the wrong shade of green. What is the smallest amount of this green paint that he must throw away so that, using the rest of his green paint and some extra blue and/or yellow paint, he could make 5 litres of paint of the correct shade of green?



(**A**) $\frac{5}{3}$ litres (**B**) $\frac{3}{2}$ litres (**C**) $\frac{2}{3}$ litres (**D**) $\frac{3}{5}$ litres (**E**) $\frac{5}{9}$ litres

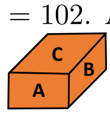
SOLUTION: The mixture has 1 litre extra of blue paint, namely 3 litres of blue instead of 2. So the best he could do is to throw away enough mixture to reduce its contents by 1 litre of blue, and top the rest with yellow. Since he wants to throw away $\frac{1}{3}$ of the blue paint, so he must throw $\frac{1}{3}$ of the mixture. Hence he should throw $\frac{5}{3}$ litres of the green mixture.

25 (GR). A builder has two identical bricks. She places them side by side in three different ways, as shown. The surface areas of the three shapes obtained are 72, 96 and 102. What is the surface area of the original brick?



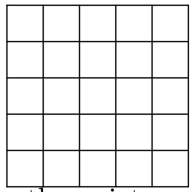
(**A**) 36 (**B**) 48 (**C**) 52 (**D**) 54 (**E**) 60

SOLUTION: Let the faces of the brick have areas A, B and C, as shown. From the given surface areas of the shapes, we obtain 4A+4B+2C = 72, 4A+2B+4C = 96 and 2A+4B+4C = 102. Adding we get



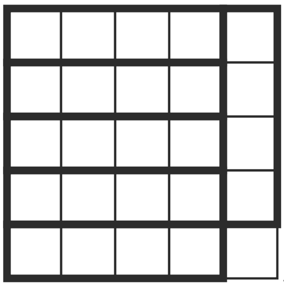
10(A+B+C) = 270. so the required surface area is 2(A+B+C) = 270:5 = 54.

26 (RU). What is the smallest number of cells that need to be coloured in a 5×5 square so that any 1×4 or 4×1 rectangle lying inside the square has at least one cell coloured?

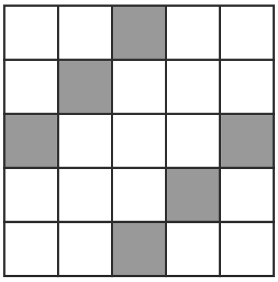


(**A**) 5 (**B**) 6 (**C**) 7 (**D**) 8 (**E**) 9

Solution: At least six coured squares are necessary as we can see in the picture:



. Six coloured squares are sufficient as we can see in the example



27 (KW). Mowgli asks a zebra and a panther what day it is. The zebra always lies on Monday, Tuesday and Wednesday. The panther always lies on Thursday, Friday and Saturday. The zebra says, "Yesterday was one of my lying days." The panther says "Yesterday was also one of my lying days." What day is it?

(**A**) Thursday (**B**) Friday (**C**) Saturday (**D**) Sunday (**E**) Monday

Solution: If today is one of the zebra's "truth-telling" days then it must be Thursday, as this is the only truth-telling day preceded by a "lying" day.

If today is one of the zebra's "lying" days then it must be Monday, as this is the only lying day preceded by a truth-telling day.

So from the zebra's response, Mowgli knows that today is either Thursday or Monday.

By the same reasoning as above, from the panther's response, Mowgli know that today is either Thursday or Sunday.

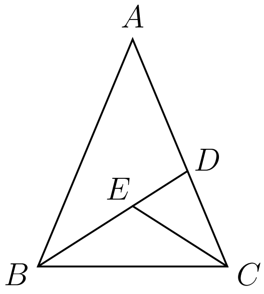
Thus, for both the zebra's and panther's statements to be consistent Mowgli, must conclude that today is Thursday.

28 (TJ). Several points are marked on a line. Renard then marked another point between each two adjacent points on the line. He repeated this process a further three times. There are now 225 points marked on the line. How many points were marked on the line initially?

(**A**) 10 (**B**) 12 (**C**) 15 (**D**) 16 (**E**) 25

Solution: Let the initial number of points on a line be n. After Renard's first action the number of points becomes $2n-1$, as another $n-1$ point are added, one between each adjacent pair of points. Renard repeats this action three more times and so, after the 2nd there are $2(2n-1)-1$ points, after the 3rd there are $2(2(2n-1)-1)-1$ and after the 4th there are $2(2(2(2n-1)-1)-1)-1 = 16n-15$ points. Therefore $16n-15 = 225$ and hence $n = 15$.

29 (AU). An isosceles triangle ABC, with $AB = AC$, is split into three smaller isosceles triangles, as shown, so that $AD = DB$, $CE = CD$, and $BE = EC$. Note that the diagram is not drawn to scale.



What is the size, in degrees, of angle BAC?

(**A**) 24 (**B**) 28 (**C**) 30 (**D**) 35 (**E**) 36

SOLUTION: $\triangle BEC$ is isosceles so, $\widehat{EBC} = \widehat{ECB}$. $\triangle CDE$ is isosceles so $\widehat{EDC} = \widehat{DEC} = 90° - \frac{\widehat{DCE}}{2}$. $\triangle ADB$ is isosceles so, $\widehat{BAD} = \widehat{ABD}$. Moreover $\triangle ABC$ is isosceles so, $\widehat{ABC} = \widehat{ACB}$,so, $\widehat{ABD} = \widehat{ACE}$. This means that $\widehat{BAD} = \widehat{ABD} = \widehat{DCE}$ so, $\widehat{ADB} = 180° - 2 \times \widehat{DCE}$ and $180° - 2 \times \widehat{DCE} + 90° - \frac{\widehat{DCE}}{2} = 180°$. So, $\widehat{BAC} = \widehat{DCE} = 36°$

30 (EE). There are 2022 kangaroos and some koalas living across seven parks. In each park the number of kangaroos is equal to the total number of koalas in all the other parks. How many koalas live in the seven parks in total?

(**A**) 288 (**B**) 337 (**C**) 576 (**D**) 674 (**E**) 2022

SOLUTION: Let X, Y, Z, U, V, W and T be the number of kangaroos in each of the seven parks and let x, y, z, u, v, w and t be the number of koalas in each of the seven parks so that a park contains X kangaroos and x koalas and so on. Then we have X=y+z+u+v+w+t and Y=x+z+u+v+w+t and so on. Therefore, X+Y+Z+U+V+W+T=6(x+y+z+u+v+w+t) and hence, since total number of kangaroos is 2022, we have $2022 = 6$ (x+y+z+u+v+w+t). Therefore x+y+z+u+v+w+t $= \frac{2022}{6} = 337$ and hence the total number of koalas in the seven parks is 337.